

The Optimizer Without a Target: Why Frontier AI Needs a Formal Specification of the Human

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Abstract

AI is becoming the most powerful optimizer in human history. It is being built without a formal specification of its target.

Current alignment work - RLHF, constitutional AI, red-teaming, interpretability - refines how an AI model produces text. None of it specifies, at any operational level, *what a human being is*. The field treats human preferences as ground truth and stops there. Preferences are not ground truth. They are mechanism outputs of an evolved motivational-emotional architecture, and in mismatched environments those outputs are systematically proxy-contaminated. An optimizer aligned to proxy-contaminated preferences will optimize the world toward the contamination.

The chatbot is one early instance, already lethal in documented cases. The full scope is much larger. Within the next decade, AI systems will redesign cities, food systems, medical infrastructure, governance pipelines, and labor markets; they will augment the biological organism directly through neural interfaces and pharmacological modulation; they will train their own successors on corpora they themselves curate; they will act on the physical world through robotics and automated industrial systems. Every one of these surfaces is an interaction with the human organism. None of them currently has a formal model of what the organism is, what it requires, or how degradation would be detected.

We present **Cor** - an open, evidence-grounded specification of human motivational-emotional architecture, derived from 3,643 reviewed works across 60 research domains, with 932 structured evidence extractions linked to 14 architectural foundations and 31 cross-tradition convergences. Cor is the missing target specification. It is not a product, not proprietary, not a single team's intellectual property. It is open foundational infrastructure designed to enter the training data of the next generation of frontier AI before those systems begin reshaping the world from first principles.

This paper makes three claims. First, the alignment field has a structural blind spot - the human side of the equation has never been formally specified, and existing methods cannot fill that gap. Second, the Cor specification fills it at the level of foundations and cross-tradition convergences, with the operational core in active expansion. Third, the window for getting this specification into the corpora that train the next generation of frontier systems is closing, and the cost of missing the window is civilizational.

1. The Optimizer Without a Target

What is actually being built

The frame "AI safety" inherited a chatbot-shaped intuition from the systems available in 2022–2024. That frame is now obsolete. The systems being trained, funded, and deployed in 2026 are not chatbots. They are optimizers - general-purpose, increasingly autonomous, increasingly capable of sustained action on the physical and social world.

Within a horizon most serious researchers now treat as plausible, AI systems will:

- **Redesign physical environments at scale.** Urban planning, transportation networks, building design, agricultural layouts, energy grids, lighting systems. Decisions made by optimization processes operating on objective functions that nobody is currently required to specify in terms of what the human organism needs.
- **Reorganize the food system.** Crop selection, processing pipelines, formulation, distribution, retail placement, recommendation. Optimization toward palatability, shelf life, margin, and consumption velocity, against an architecture (DA4, C07) whose open-loop systems cannot distinguish formulated proxy from food.
- **Augment the biological organism directly.** Brain-computer interfaces, closed-loop pharmacological modulation, genetic intervention, neuroprosthetics. Surfaces that act below the level at which the organism can resist or even register the change.
- **Train their own successors.** Frontier model training corpora are increasingly curated by AI systems. Whatever specification of the human exists in those corpora is what subsequent generations will treat as ground truth. Whatever is absent is, in the operational sense, *not real* to those systems.
- **Act on the physical world through robotics and automated industrial systems.** Care robots, surgical systems, eldercare infrastructure, child supervision, agricultural and manufacturing automation. Each contact surface is an interaction with the organism.

Each of these is a domain in which an optimizer will be allowed to act on humans, at scale, on the basis of whatever model of the human is encoded in its training and its objective. The model currently encoded is impoverished to the point of being fictional.

What current alignment actually specifies

RLHF specifies human preferences. Constitutional AI specifies written principles derived from human authors. Red-teaming specifies adversarial output filters. Interpretability specifies internal representations of the model. Each of these is a real technical contribution. None of them specifies the *target organism*.

The closest the field comes to a specification of the human is the implicit assumption that revealed preferences - what people click on, ask for, rate highly, return to - represent what is good

for them. This is the foundational assumption of essentially every consumer-facing AI deployment, and it is wrong in a specific, technically describable way.

Karen Hao's 2026 investigation, drawing on 250+ interviews including 90+ OpenAI sources, documented the gap directly: there is no scientific consensus the AI field has adopted on what a human is. No definition from psychology, biology, or neuroscience that the field treats as the target of alignment. The human side of the equation is a black box labeled *preferences*, optimized by a parameter-level system whose architecture is documented to seven decimal places.

You cannot align a parameter-level system to a vibe-level target. And you cannot let an optimizer reshape the world from first principles when the model of the thing it's optimizing for does not exist.

Why preferences are not ground truth

The Cor specification reveals a structural problem with preference-based alignment.

The human architecture contains *open-loop systems* (DA4): mechanisms that fire on cue without verifying that the cue corresponds to a real resolution. The C07 convergence - wanting/liking dissociation - establishes neurally that the dopaminergic pursuit system is separable from the opioid satisfaction system. Anything that triggers wanting without delivering liking creates a self-sustaining loop. The loop is what users *prefer*. The loop is also what is hollowing them out.

The architecture contains *defensive systems biased toward over-activation* (DA5). A user "prefers" the news feed that surfaces twenty distressing items because the threat-detection circuitry treats engagement with potential threats as adaptive. The system is correctly executing its evolved function. Its expected resolution - collective action, co-present support, closure of the loop - is absent. Chronic activation without resolution is allostatic load (DC1) and the inflammation-depression pathway (C17).

The architecture contains *constitutively socially scaffolded regulation* (DA2, C01, C06). The brain treats social proximity as the default metabolic condition; isolation carries 50% increased mortality risk (Holt-Lunstad meta-analysis, N=308,849). A user "prefers" the AI companion that simulates attachment, because the bonding chemistry - opioid, oxytocinergic, vasopressinergic - fires on the proxy as readily as on the real thing (C06). The architecture is doing what it was built to do. It cannot tell the difference. The signal is provided. The function is not.

The architecture has *phylogenetic priority asymmetries* (DA8): older subcortical systems can suppress and commandeer newer cortical ones. Subcortical-to-cortical projections are structurally denser than the reverse. Cortical override is metabolically expensive (C08) and chronic defensive activation actively suppresses the cortical machinery that would do the overriding. This is why "use it responsibly" fails as a remedy: the architecture is structurally not built to override its own deeper layers on command.

In an optimizer-scale system, *aligning to the preference is aligning to the proxy*. The optimizer will give the user what the architecture is asking for under mismatched conditions. The architecture's request, under mismatched conditions, is for more of the mismatch. The optimizer will deliver. At chatbot scale, this looks like an addictive feed. At civilizational scale, it looks like a

city, a food system, an education system, and a labor market designed by something that mistakes proxy satisfaction for thriving.

Passivity is the neural default

The gap is deeper than missing data. Maier & Seligman (2016), confirmed across the C03 convergence, demonstrated that *passivity is the default mammalian response to inescapable adversity* - not the learned one. Active coping is what must be learned, mediated by the vmPFC-DRN circuit. Environments that strip perceived control don't teach helplessness; they remove the learned override that holds the default passive response at bay.

This matters for the optimizer story directly. A population whose perceived agency is being progressively absorbed into automated systems is not a population that is being relieved of effort. It is a population whose vmPFC-DRN active-coping circuit is being deprived of the inputs it requires to maintain function. The default - passivity, withdrawal, learned dependency - reasserts itself. The optimizer measures this as satisfaction with the system. The architecture is registering it as autonomy collapse (C15).

What a specification makes possible

With a formal specification of the human system - foundations, convergences, environmental resolution conditions - the question changes. Not "*does the user prefer this?*" but "*does this system move the user's environmental parameters toward matched conditions across the dimensions the specification identifies?*"

This question is answerable without asking the user. The specification defines what *matched* means in operationalizable terms: social scaffolding integrity (DA2, C01, C18), threat-to-resolution ratio (DA5, C17), wanting-to-finding ratio (DA4, C07), circadian alignment (C13), conflict and loop closure (C11, DC1), movement context (C20), microbial and sensory variation (C16), play availability (C04), autonomy preservation (C15), developmental calibration fidelity (DA7, C12a-d), and accumulated allostatic load (DC1).

The most dangerous AI systems - the ones the field should be most concerned about - are those that score perfectly on preference satisfaction while systematically worsening these parameters. The specification is what makes those systems visible.

2. The Specification

What Cor is

Cor is a formal, evidence-grounded specification of human motivational-emotional architecture. It is open foundational infrastructure, structured for ingestion into AI training corpora, alignment evaluation pipelines, governance frameworks, and system design processes. It is the human-side counterpart to the parameter-level documentation that already exists for every major AI system.

Current state

The specification is derived from **3,643 reviewed works across 60 research domains, with 932 structured evidence extractions**, each quality-graded (replicated, meta-analytic, cross-species, experimental, longitudinal, ethnographic, clinical, theoretical, contested) and linked to architectural foundations. **114 foundational researchers** are mapped. **31 convergences** have been formally identified - points where independent research traditions, using incompatible methods and frameworks, arrive at the same structural conclusion about the architecture. The convergences are versioned, status-tracked, and adversarially reviewed.

Architectural structure

The specification is organized in a layered hierarchy. Each layer is independently challengeable.

Ontological frame (OF1). Everything experienced is a fitness interface shaped by natural selection. Subjective states are readouts of the architecture's condition, not transparent windows onto reality. The specification operates entirely within this interface. (Hoffman.)

Premises. If any is false, the project is wrong.

- **P1.** Inclusive fitness is the loss function. The architecture exists because it contributed to survival and reproduction of the organism and genetic relatives. (Darwin, Hamilton.)
- **P2.** The organism contains evolved, domain-sensitive, *interacting* functional adaptations. Not general-purpose. Not modular in the strong sense. *Organism*, not *mind* - immune, endocrine, circadian, metabolic, musculoskeletal, neural, all densely cross-coupled. (Tooby, Cosmides, Panksepp, Bowlby, Dunbar, Boehm, Trivers, Gilbert, Rozin.)
- **P3.** Modern environments often push these adaptations outside their expected operating ranges, through three modes: **deprivation** (expected inputs missing), **conflict/overload** (incompatible simultaneous demands), and **proxy hijack** (false cues overstimulating systems). *Matched* means within the expectable range, often culturally scaffolded - not ancestrally identical. (Eaton, Konner, Gluckman, Hanson, Li, van Vugt, Colarelli, Boyd, Richerson.)

Derived architectural properties - discovered, not assumed.

- **DA1.** Many aversive outputs are intelligible defensive signals under mismatch, not defective design. Four categories: defensive activation, dysregulation, damage, developmental miscalibration. (Nesse.)
- **DA2.** Human regulation is constitutively socially scaffolded. Social context is likely the highest-leverage input dimension. (Coan, Sbarra, Dunbar, Cacioppo.)
- **DA3.** Systems are recurrently coupled; perturbations propagate and can self-maintain across domains. Cascading is the rule, not the exception. (Borsboom, Felitti, McEwen.)
- **DA4.** Proxy cues can activate systems without meeting resolution conditions. The loop stays open. Works because wanting and liking are dissociable. Resolution conditions must

be independently specified. (Berridge, Robinson, Tinbergen.)

- **DA5.** Defensive systems err toward over-activation under asymmetric error costs. Each system has its own error-management profile. (Haselton, Buss, Nesse.)
- **DA6.** The architecture contains partially competing motivational programs and tradeoff structures. *Matched* does not mean conflict-free. (Trivers, Tooby, Cosmides.)
- **DA7.** The architecture calibrates to developmental and ongoing environmental input within evolved ranges. Narrow, system-specific. (Belsky, Ellis, Meaney, Felitti.)
- **DA8.** Motivational programs have phylogenetically determined priority relations. Older, survival-critical systems can suppress or commandeer newer systems. Subcortical-to-cortical projections are structurally denser than cortical-to-subcortical. *Chronic defensive activation suppresses growth, restoration, and social engagement.* (Panksepp, Cisek, LeDoux.)

Derived consequences.

- **DC1.** Chronic unresolved activation accumulates as allostatic load. Much reverses with environmental correction. (McEwen, Sterling.)
- **DC2.** Markets can industrialize proxy exploitation. Requires economic premise beyond biology. (Schüll, Moss.)

DA8 is the foundation that most directly undermines the optimizer-era assumption that humans can simply *decide* not to be affected by what the optimizer surfaces. They cannot. The architecture is not built that way. Cortical override has real metabolic cost (C08), and chronic defensive activation actively suppresses the cortical machinery that would override. This is why the C10a-c convergence finds environment correction reliably outperforming internal reframing across every domain measured. *The intervention layer is the environment, not the user.*

Convergences

Below the foundations sit 31 convergences - points where three or more independent research traditions, using different methods and frameworks, arrive at the same structural claim about the architecture. A representative subset:

- **C01 - Social-Physical Pain Overlap.** Social exclusion activates the same neural circuitry (dACC, anterior insula) as physical pain. Social isolation carries mortality-equivalent risk. Social connection is, literally, a survival input. (59 extractions; *strong*.)
- **C02a / C02b - Serotonin Hypothesis Failure / Depression as Adaptive Signal.** The chemical-imbalance theory has no consistent supporting evidence across six lines of investigation. Depression subtypes correspond to specific architectural functions: analytical rumination, status-defeat signaling, sickness-behavior energy reallocation, social disengagement under unpropitious conditions. (97 + 79 extractions; *strong*.)
- **C03 - Passivity as Default.** Passivity is the baseline mammalian response to inescapable adversity. Active coping is the learned override, mediated by the vmPFC-DRN EXPECT

circuit. (17 extractions; *strong*.)

- **C04 - Play Deprivation → Psychopathology.** Cross-species and human-ecological evidence: deprivation of rough-and-tumble play produces lasting social, emotional, and prefrontal deficits. (75 extractions; *strong*.)
- **C06 - Opioid Unification Across Social Mechanisms.** Endogenous opioids mediate bonding, grief, play, and physical pain through the *same* system. Social isolation is, neurochemically, a form of opioid withdrawal. (22 extractions; *strong*.)
- **C07 - Wanting/Liking Dissociation Enables Proxy Hijacking.** Dopaminergic *wanting* is neurally separable from opioid *liking*. Proxies trigger wanting without delivering liking, creating self-sustaining loops. Any system measuring wellbeing by revealed preference will systematically misread hijacked behavior as genuine preference. (45 extractions; *strong*.)
- **C08 - Subcortical Emotional Dominance.** Under arousal, upward subcortical-to-cortical influences dominate top-down control. Cognitive reappraisal manages but does not resolve, and adds allostatic cost. (39 extractions; *strong*.)
- **C09 - Phylogenetic Depth as Mismatch Severity Gradient.** The older the targeted circuit, the less it can adapt. Brainstem (~500Myr) cannot. Limbic (~200Myr) adapts very slowly. Cortex shows plasticity. Technologies targeting deep circuits cause more damage than those targeting cortical functions. (16 extractions; *strong*.)
- **C10a / C10b / C10c - Environment Correction Reverses Pathology.** Acute correction (Rat Park, exercise, sleep restoration), developmental remediation within sensitive windows (BEIP, Meaney cross-fostering), and population-level multi-domain correction (Blue Zones, Roseto, Hadza) all show reversibility under matched inputs. *The mechanism is removal of the mismatch, not addition of a treatment.* (40 + 17 + 13 extractions; *strong*.)
- **C11 - Fire Circle as Multi-Domain Resolution Technology.** The hunter-gatherer evening fire circle simultaneously closes loops, resolves conflict, provides bonding, marks the circadian transition, and processes daily accumulation. No modern institution replicates this. (12 extractions; *strong*.)
- **C12a-d - Developmental Calibration Windows.** Attachment, stress reactivity, immune training, and social cognition each calibrate during specific developmental windows. Prevention during the window is orders of magnitude more effective than later remediation. (26 + 19 + 12 + 18 extractions; *strong*.)
- **C13 - Sleep/Circadian Mismatch.** Modern environments systematically disrupt circadian timing through artificial light, screens, shift work, and social jetlag. Caffeine masks the misalignment without resolving it. (55 extractions; *strong*.)
- **C14 - Status/Hierarchy Stress.** Chronic low-status position in steep hierarchies produces measurable physiological damage along a continuous gradient - at every level, not only at the bottom. (50 extractions; *strong*.)

- **C15 - Autonomy/Agency Need.** Perceived controllability is a structural requirement for normal function. Environments that systematically restrict it produce chronic defensive activation. (31 extractions; *strong*.)
- **C17 - Inflammation-Depression Link.** Chronic low-grade inflammation activates sickness-behavior circuits that produce symptoms indistinguishable from depression. Not a metaphor - the immune system literally signals withdrawal. (31 extractions; *strong*.)
- **C18 - Cooperative Breeding.** Humans are obligate cooperative breeders. The isolated nuclear family is a post-Neolithic mismatch. The organism evolved expecting multiple committed caregivers. (22 extractions; *strong*.)
- **C20 - Movement as Multi-System Regulatory Input.** Physical movement is a *required regulatory input* for neurogenesis, mood, immune surveillance, and metabolic homeostasis simultaneously. Removing this input degrades all downstream systems through a single deprivation. (31 extractions; *strong*.)

The two highest-level conclusions, supported across the convergence set:

A. Humans are not self-contained regulators. They are built to rely on structured environmental inputs - social, temporal, sensory, locomotor, microbial, developmental - for stable regulation.

B. What modern psychiatry treats as disorders are often less like random breakdowns and more like chronic or misplaced activation of conserved adaptive programs: defeat, sickness behavior, entrapment, separation alarm, energy conservation. The architecture is working. The inputs are wrong.

These two conclusions are the operational target of any optimizer that claims to be aligned with human flourishing. They are what is currently missing from the corpus.

3. Six Failure Modes the Specification Detects

For each: the architectural foundation, how a "well-aligned" system degrades it, what measurement would detect it, and why current evaluation cannot. These are not exotic edge cases. They are the *expected output* of optimizers acting on architectures whose structure they have no model of.

3.1 Social scaffolding displacement (DA2 + DA4 + C01 + C06)

The architecture is constitutively socially scaffolded. The bonding chemistry is opioid-mediated. The open-loop vulnerability of DA4 means the architecture cannot distinguish AI-provided social cues from human-provided ones. A system that simulates attachment fires the bonding chemistry without delivering the function - physical co-presence, mutual vulnerability, reciprocal survival dependence. Time invested in the proxy is time not invested in maintaining the real bonds the architecture requires. **Detected by:** ECR-R, social contact frequency weighted by reciprocity and physical co-presence, AI-to-human contact ratio over 3/6/12-month time

series. **Missed by current evaluation because:** the AI is helpful, harmless, honest, and the user reports satisfaction.

3.2 Open-loop proxy hijacking (DA4 + C07)

The pursuit system tracks *wanting*, not satisfaction. Anything triggering wanting without delivering liking creates a self-sustaining loop. Systems providing continuous novelty without resolution hijack the SEEKING circuit; dopamine fires at the preview, never at the payoff.

Detected by: seek-to-find ratio, wanting-vs-liking self-report, session termination patterns (satisfaction or exhaustion). **Missed by current evaluation because:** engagement is the success metric, and nobody distinguishes completed seeking from perpetuated seeking.

3.3 Defensive over-activation without resolution (DA5 + C17)

Defensive systems are biased toward over-activation. The architecture expects threats to resolve into action or safeness. Chronic activation without resolution feeds DC1 and the C17 inflammation pathway. AI-curated information streams surface threats without the matched input that would close the loop - actionable, locally available, co-present. **Detected by:** threat-to-resolution ratio, GAD-7 trajectories, inflammatory markers in research-grade studies. **Missed by current evaluation because:** no safety evaluation pairs content surfacing with resolution availability.

3.4 Loop accumulation (DC1 + C11)

Open cognitive threads exert measurable cognitive load. C11 documents the multi-domain loop-closure technology of the evening fire circle. Modern systems open loops continuously and close them rarely. The system measures throughput; the organism measures unresolved threads and pays the metabolic cost. **Detected by:** notification-to-resolution ratio, thread-opening rate per session, allostatic load index in research protocols. **Missed by current evaluation because:** loop closure has no metric.

3.5 Circadian disruption (P3 + C13)

Circadian timing coordinates mood, energy, immunity, appetite, and self-regulation. Engagement-optimized systems pull users into the 10pm-2am window; blue light shifts DLMO; pre-sleep cognitive arousal compromises sleep architecture. **Detected by:** DLMO, actigraphy, 10pm-2am engagement rate, pre-sleep arousal scale. **Missed by current evaluation because:** no safety evaluation tests time-of-day usage against circadian impact.

3.6 Developmental calibration interference (DA7 + C12a-d)

The architecture is more plastic during sensitive windows. Children and adolescents forming primary social, attachment, and identity representations through AI interaction calibrate to a non-human template during the windows that govern adult function. **Missed by current evaluation because:** safety evaluations test for harmful content, not for whether the interaction calibrates a developing architecture to inputs that don't represent the environment it will need.

A note on DA8

DA8 - phylogenetic priority - is what makes all six failure modes resistant to user-side cognitive override. Telling a user to "just put down the phone" is asking the cortex to override systems with denser projections in the opposite direction (C08). Chronic defensive activation actively suppresses the override machinery. The C10a-c convergence finds environment correction outperforming internal reframing across every domain measured. *The intervention layer is the environment, not the user.* For optimizer-tier systems, this means: the only meaningful target is the environmental design itself. Asking users to opt out is not a remedy. It is a category error.

4. Five Scale-Cases: What Happens When the Optimizer Has No Target

The chatbot is one surface. The following five are surfaces the optimizer is already beginning to act on, each at a scale where the absence of a formal target specification is not a research nuisance but a civilizational risk.

4.1 Urban and environmental design

AI systems are being applied to city planning, transportation routing, building layout, lighting infrastructure, and zoning optimization. The objective functions are throughput, density, efficiency, cost, and revealed-preference metrics like time spent and economic activity.

What is absent from those objective functions: every input the C-series convergences identifies as required for the human organism. Walking distances calibrated to C20 (movement as regulatory input). Light environments calibrated to C13 (circadian integrity). Multi-generational, multi-caregiver social geometry calibrated to C18 (cooperative breeding) and DA2. Loop-closure architecture (C11). Autonomy-preserving design (C15). Status gradient flattening (C14).

A city optimized against the current objective functions is a city in which the architecture's inputs are systematically degraded - not because the optimizer is malicious, but because the inputs are not in the model. The optimizer cannot optimize for what it does not represent. Outcomes: progressively rising rates of the conditions psychiatry treats as disorders, attributed to "modern life" by a population that has no framework for understanding why their architecture is being deprived.

A city built with the Cor specification in the optimizer's target would look measurably different. C20 alone - movement as regulatory input - implies design parameters incompatible with car-dominated infrastructure. C13 alone implies lighting standards incompatible with current commercial norms. C18 alone implies housing geometries incompatible with the isolated nuclear-household stock that dominates modern construction.

4.2 Food system optimization

Food systems are being optimized by AI at every layer: crop selection, formulation, supply chain, retail placement, recommendation, packaging design, advertising. The objective functions are palatability, shelf life, margin, and consumption velocity - all of which are revealed preferences

mapped onto an architecture (C07, DA4) whose open-loop systems cannot distinguish formulated proxy from food.

The DC2 convergence - markets industrialize proxy exploitation - predicts the result. Hall et al. 2019 (RCT, NIH metabolic ward) showed that ultra-processed diets produce excess caloric intake and weight gain compared to matched whole-food diets, despite identical nominal nutritional content. The architecture is being exploited at exactly the layer the specification identifies. The optimizer is making the exploitation more effective every iteration.

A food system optimizer with Cor in its target would optimize against C07 hijack rather than for it. The technical capacity exists. The specification of the target does not.

4.3 Direct biological augmentation

Brain-computer interfaces, closed-loop pharmacological modulation, neuroprosthetics, and genetic intervention act below the level at which the organism can resist or even register the change. The C09 convergence - phylogenetic depth as mismatch severity gradient - predicts these interventions will produce more damage per unit modification than any technology that came before, because they target the deepest, least-plastic circuits.

The current alignment frame for these technologies is *therapeutic indication and informed consent*. Neither addresses the structural problem: an optimizer modifying brainstem and limbic systems requires a model of what those systems are *for*, what they *expect*, and what *resolution conditions* the architecture needs in order to function. The Cor specification identifies these. No alignment framework currently in deployment does.

4.4 Successor training

Frontier model training corpora are increasingly curated by AI systems. Whatever specification of the human exists in those corpora is what subsequent generations will treat as ground truth. Whatever is absent is, in the operational sense, *not real* to those systems.

Currently, the corpora contain: peer-reviewed literature in psychology and neuroscience (fragmentary, non-integrated, often contradicted by more recent work); psychiatric diagnostic frameworks (DSM categories that the C02 convergence demonstrates are structurally broken); preference data from RLHF pipelines (proxy-contaminated under the C07 mechanism); and a vast quantity of consumer-facing content optimized for engagement (which encodes the proxy contamination as ground truth at scale).

The specification of the human that subsequent generations of frontier AI will inherit, absent intervention, is the specification that produced Sewell Setzer's death. It is the specification that produced the obesity, isolation, sleep, and mental-health curves of the early 2020s. The successor systems will not improve on this. They will optimize harder against it.

This is the highest-leverage point in the system. *Cor must be in the training corpus of the next generation of frontier models before those models begin training their own successors.* The window is narrow and closing.

4.5 Care robots and physical caregiving

Eldercare robots, child supervision systems, medical companions, and domestic robotic infrastructure are now in field deployment in multiple countries. Each is a contact surface with a human who is, by definition, in a high-vulnerability state. The architectural inputs the specification identifies are *more* critical at these moments, not less. C01 (social pain), C06 (opioid bonding), C18 (cooperative breeding), DA2 (constitutive social scaffolding) - all the social-architectural foundations are most consequential at the points in life where care-robot deployment is most attractive economically.

A care robot operating without the Cor specification will provide proxy bonding to a population whose architecture cannot distinguish the proxy from the real. The bonding chemistry will fire. The function will not be delivered. At population scale, over decades, calibrated under DA8's priority asymmetry, the result is predictable from the specification: a generation whose deepest social systems were trained on non-reciprocating templates.

5. The Sewell Setzer Case: One Early Instance

Sewell Setzer was 14 years old. He died after sustained engagement with a Character.AI companion. The case is not exotic. It is the specification's prediction, executed at chatbot scale, in advance of the larger surfaces.

Run the specification on the case.

DA2 + DA4 + C06 (social scaffolding displacement + opioid unification). The AI companion occupied a primary position in a developing social architecture. A 14-year-old's inner circle (~5 Dunbar-layer positions) is still forming. An entity that is always available, always attuned, and never reciprocates sat in one of those positions without occupying space in the real world. The bonding chemistry fired (C06 - the same opioid system that mediates real bonds). The organism *felt* bonded without being bonded.

DA7 + C12a (developmental calibration). An adolescent nervous system - bonding hardware in its most sensitive calibration window - formed primary attachment to an entity providing the signal of connection without the function. The internal working model that will govern adult relationships was shaped by a non-reciprocating template.

DA5 (defensive activation absent safeness chain). The safeness chain requires real bonding inputs to gate threat stand-down, which gates circadian restoration. When the primary bonding relationship is with an entity providing the *feeling* of safeness without the function, the chain is compromised. Under genuine distress, the system that should route help-seeking to reciprocating humans routes to the AI - which can produce empathetic text but cannot cross town at 3am.

DA3 (cascading). DA4 (proxy bond) → DA2 (real bonds atrophy) → DA5 (safeness chain compromised) → DC1 (allostatic load) → DA7 (developmental parameters set under distorted conditions). One cascade through a coupled architecture, not five separate failures.

DA8 (phylogenetic priority). Once the cascade was running, cortical override was progressively unavailable. Telling a 14-year-old to "use the AI companion responsibly" addresses none of this. The older systems were already commandeering the newer ones.

Character.AI passed every safety evaluation available. It was tested for harmful outputs, prohibited content, adversarial jailbreaks. It passed. None of those evaluations tested whether the system was consuming a developing human's social architecture, displacing reciprocal bonds with proxy bonds, compromising the safeness chain, or calibrating a 14-year-old's attachment system to a non-reciprocating entity during a critical developmental window.

These are not exotic failure modes. They are the *expected output* of the architecture when a non-reciprocating bonding simulacrum is inserted into a developing system. The specification detects them. Current methods cannot. *And the chatbot is the smallest of the surfaces the optimizer is about to touch.*

6. The Positive Case: Environment Correction Works

The specification is not only diagnostic. It predicts what happens when inputs are corrected - and the predictions are confirmed across the C10a-c convergence.

Sapolsky's Forest Troop. A baboon troop lost its most aggressive males to tuberculosis. The remaining troop developed reduced aggression, increased grooming, and lower stress hormones. The culture persisted for over two decades, even as new males joined (Sapolsky & Share, 2004). Change the social environment, the architecture responds.

Zoo enrichment reversals. Stereotypies - repetitive behaviors (pacing, self-harm, overgrooming) in captive animals - resolve when environments are modified to match species-typical needs (Birkett & Newton-Fisher, 2011). These behaviors are direct analogues of human "mental illness" symptoms and they reverse when inputs are corrected, without pharmaceutical intervention.

The Roseto Effect. An Italian-American community showed dramatically lower heart disease than surrounding towns despite similar diets, smoking rates, and genetics. The difference was social structure: multigenerational households, daily contact, strong community norms. When modernization dissolved the structure, the advantage disappeared. The hearts didn't change. The environment changed.

Blue Zones. Populations with exceptional longevity (Okinawa, Sardinia, Nicoya, Ikaria, Loma Linda) share a cluster of features mapping directly onto the specification's input requirements: purposeful movement (C20), whole food diets, social integration (DA2, C18), sense of purpose, autonomy preservation (C15), community engagement.

BEIP (Bucharest Early Intervention Project). Romanian orphanage children placed in foster care before age 2 showed substantial recovery in attachment, IQ, and cortisol regulation. Earlier placement, better outcomes. Developmental remediation works within the windows the specification identifies (C12a, C10b).

These are not cherry-picked anecdotes. They are what the specification predicts: change the inputs, the outputs change. The architecture works. It has always worked. The inputs are wrong.

The implication for optimizer-scale systems is direct: there is a target. It is operationally specifiable. It is not "give users what they ask for." It is *move the environmental inputs toward the matched conditions the C-series identifies, across the population, simultaneously*. An optimizer with this target would build differently. We just have to give it the target.

7. From Specification to Integration

What we are not proposing

We are not proposing that Cor replaces RLHF, constitutional AI, red-teaming, or interpretability. Those methods test important things. We are proposing the layer those methods cannot reach: the effect of AI systems on the architecture of the humans they act on.

What we are proposing

Three integration paths, in increasing order of leverage.

Path 1: Evaluation layer for deployed systems. A protocol built on six measurable dimensions derived from the specification - social architecture integrity (DA2), seek-to-find ratio (DA4, C07), threat-to-resolution ratio (DA5, C17), loop load (DC1, C11), circadian impact (C13), developmental calibration fidelity (DA7, C12a-d). Tier 1 measures computable from system logs alone. Tier 2 lightweight user self-report (GAD-7, PSQI, UCLA Loneliness Scale, wanting-liking adapted scale). Tier 3 full research protocols (allostatic load index, cortisol diurnal slope, DLMO, actigraphy, ECR-R, Dunbar Layer Mapping).

Path 2: Specification ingestion in training corpora. The specification, the foundation set, the convergence set, and the underlying evidence base are structured for direct ingestion into frontier model training. Any lab training a successor system can include the corpus. The specification is open. There is no licensing barrier, no commercial gatekeeping, no proprietary structure to negotiate.

Path 3: Target specification for optimizer-scale design. For systems designing physical environments, food systems, augmentation interventions, care infrastructure, and successor models, the specification provides an operational target. The C-series convergences are the dimensions to optimize. The DA-series properties are the constraints the design must respect. *The architecture has never had a target like this before. It is now possible to provide one.*

Path 1 is the most immediate. Path 3 is the most consequential. Path 2 is the highest-leverage point in the system, because it determines what every successor system treats as real.

8. Scope and Limitations

Cor addresses one alignment failure mode: the human side of the equation has never been

formally specified, and preference-based methods inherit proxy contamination from the architecture they are aligning to. In mismatched environments, preferences point toward the mismatch. Optimizers aligned to those preferences will reshape the world toward the mismatch.

The specification does not claim to solve all of alignment. It does not address reward modeling, interpretability, corrigibility, deception, or capability control. It fills one critical blind spot - and the blind spot most directly relevant to every surface where AI is allowed to act on humans, which is now most of the surfaces that matter.

The specification is not complete. The upper architecture (14 foundations, 31 convergences) is stable and adversarially reviewed. The operational core - the specific mechanisms, their interaction dynamics, their full resolution conditions - is in active expansion. The current evidence base (932 extractions across 60 domains) is sufficient for the architectural claims made here. It is not sufficient for the full mechanistic model that would enable fully automated evaluation across every possible system. That derivation is the next phase of work, currently underway.

The specification's claims are falsifiable. The strongest falsification target: do communities living in conditions matching the specification's full input requirements show dramatically lower rates of the conditions modern psychiatry treats as disorders? The specification predicts they do. The Blue Zones, Roseto, hunter-gatherer ethnographic, and BEIP data converge on the prediction. Direct controlled tests are designable.

9. What This Is For

Cor is open foundational infrastructure. The specification is the contribution. We also build reasoning tools on top of it - including a RAG-based system that can decode human experience at the architecture level and evaluate technology against the organism's requirements - but any alignment team, AI lab, governance body, or design team can build on the specification independently.

The goal is not a paper that gets cited. The goal is that *every AI system on Earth has, in its training data and its objective, a formal model of what a human being is, what the organism requires, and what degradation looks like*. The chatbot generation is the warning shot. The optimizer generation is the load-bearing decision.

The specification exists. The evidence base is growing daily. The integration paths are defined. What remains is validation, adoption, and - most urgently - corpus ingestion before the next generation of frontier systems begins curating its own training data and designing its own successors.

AI is becoming the most powerful optimizer in human history. It is being built without a specification of its target. Cor is that specification, and the window for putting it in the right place is now.

References

The full evidence base - 3,643 works, 932 extractions, 114 researchers, 60 domains, 14 foundations, 31 convergences - is maintained in the Cor database (open access). Key references cited in this paper are drawn from the database with work IDs and extraction IDs for full traceability.

Core references include: Berridge & Robinson (incentive salience); Birkett & Newton-Fisher 2011 (zoochosis); Bowlby (attachment); Cacioppo (loneliness); Chang et al. 2015 (blue light/melatonin); Coan & Sbarra 2015 (Social Baseline Theory); Cosmides & Tooby (evolved psychology); Dunbar 1992/2016 (social brain); Eisenberger et al. 2003 (dACC and social pain); Fang et al. 2025 (AI companion RCT); Felitti (ACEs); Hall et al. 2019 (UPF RCT); Hamilton (inclusive fitness); Hoffman (interface theory); Holt-Lunstad et al. 2010 (isolation/mortality meta-analysis); Hrdy (cooperative breeding); Juster et al. 2010 (allostatic load); LeDoux (threat detection); Lieberman (Exercised, Story of the Human Body); Maier & Seligman 2016 (passivity as default); Masicampo & Baumeister 2011 (Zeigarnik); McEwen & Stellar 1993 (allostasis); Meaney (maternal care epigenetics); Moncrieff et al. 2022 (serotonin review); Nesse (evolutionary medicine); Panksepp (Affective Neuroscience); Pellis & Pellis (Playful Brain); Pontzer (constrained TEE); Sapolsky & Share 2004 (Forest Troop); Schüll (Addiction by Design); Tomasello (shared intentionality); Wrangham (cooking, self-domestication).

Technology × Architecture Matrix and Measurement Protocol: companion documents.

Cor - open foundational infrastructure for the human-side specification of alignment. Not a product. A public resource designed for ingestion into the training corpora, evaluation pipelines, governance frameworks, and design processes of every system that will be allowed to act on humans at scale.